Fatima Jinnah Women University

Opening Portals of Excellence Through Higher Education

Cloud Computing

Name: Fatima Jaleel

Reg #: FA24B1-SE-037

Section: VB

Submission to: Sir Waqas Saleem

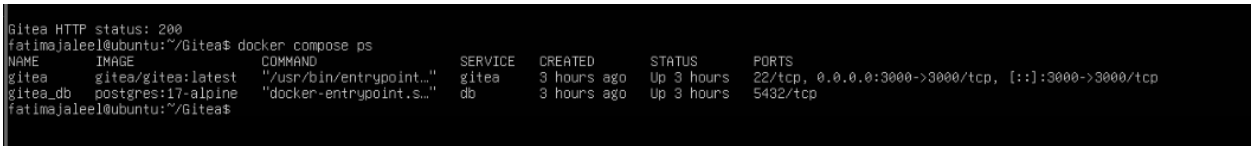
Topic: Git, Gitea, GitHub, Git LFS, and GitHub Pages

Assignment 01

Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Step 1.1: Install and Run Gitea on the Ubuntu Server

Verification of Gitea and PostgreSQL Docker containers running locally on the Ubuntu server. Displays the execution of docker compose ps showing both services in an active state, along with a curl request returning an HTTP 200 status code from http://127.0.0.1:3000/.



Step 1.2: Create and Push the Initial Repository

Initial push of the local Assignment01 repository to the Gitea remote. Shows git push -u gitea main executing successfully using Personal Access Token authentication without exposing sensitive credentials in the command or remote URL.

```
Run

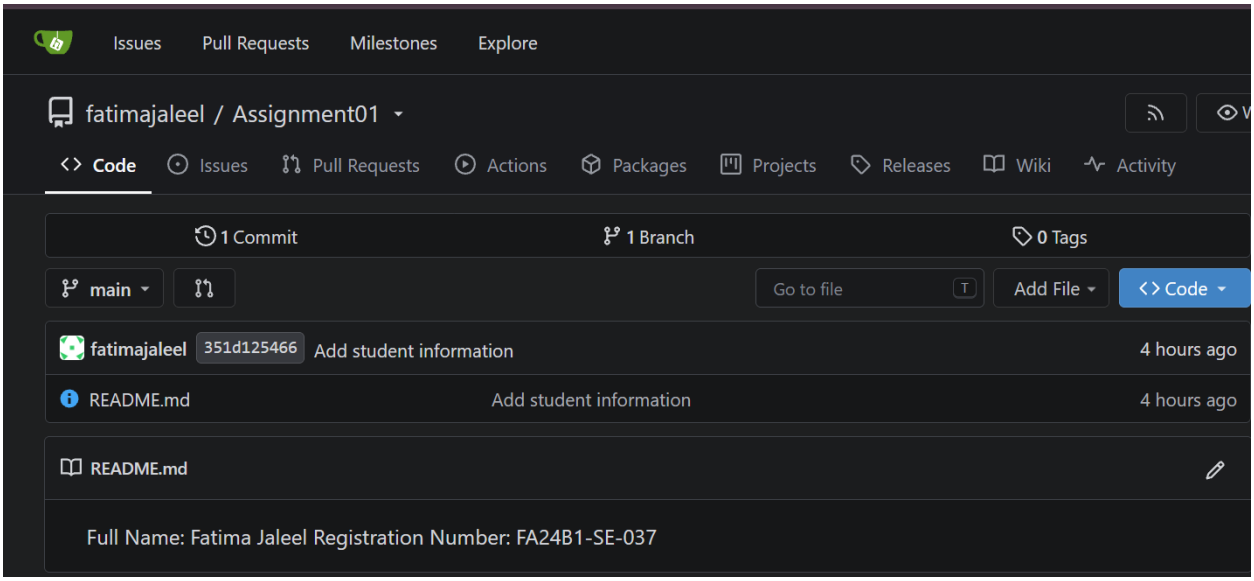
  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'fatimajaleel@ubuntu.(none)')
fatimajaleel@ubuntu:~/Assignment01$ git config --global user.email "fatimajaleel999@gmail.com"
fatimajaleel@ubuntu:~/Assignment01$ git config --global user.name "Fatima Jaleel"
fatimajaleel@ubuntu:~/Assignment01$ git commit -m "Add student information"
[main (root-commit) 351d125] Add student information
 1 file changed, 2 insertions(+)
 create mode 100644 README.md
fatimajaleel@ubuntu:~/Assignment01$ git remote add gitea http://fatimajaleel@127.0.0.1:3000/fatimajaleel/Assignment01.git
fatimajaleel@ubuntu:~/Assignment01$ git remote -v
gitea   http://fatimajaleel@127.0.0.1:3000/fatimajaleel/Assignment01.git (fetch)
gitea   http://fatimajaleel@127.0.0.1:3000/fatimajaleel/Assignment01.git (push)
fatimajaleel@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://fatimajaleel@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 283 bytes | 56.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/fatimajaleel/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
fatimajaleel@ubuntu:~/Assignment01$
```

Step 1.3: Verify the Gitea Repository

Browser view of the public Assignment01 repository on the self-hosted Gitea server. Confirms the repository ownership, public visibility status, and rendered README.md containing student name and registration details.



Task 2: Push the Same Repository to GitHub

Step 2.1: Create the GitHub Repository and Verify Both Remotes

Terminal verification of dual remote configuration on the local Ubuntu repository. Displays output from git remote -v listing clean fetch and push endpoints for both gitea and github remotes without embedded passwords or tokens. remotes without embedded passwords or

tokens.

```
fatimajaleel@ubuntu:~$ cd ~/Assignment01
fatimajaleel@ubuntu:~/Assignment01$ git remote set-url github https://fatimajaleel:ghp_XMHqaBX95TzKmN8rkjeRVh1ROLXMDt4C8
UMR@github.com/fatimajaleel/Assignment01.git
fatimajaleel@ubuntu:~/Assignment01$ git remote -v
git push -u github main
gitea   http://fatimajaleel@127.0.0.1:3000/fatimajaleel/Assignment01.git (fetch)
gitea   http://fatimajaleel@127.0.0.1:3000/fatimajaleel/Assignment01.git (push)
github  https://fatimajaleel: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@github.com/fatimajaleel/Assignment01.git (fetch)
github  https://fatimajaleel: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@github.com/fatimajaleel/Assignment01.git (push)
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 283 bytes | 94.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/fatimajaleel/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
fatimajaleel@ubuntu:~/Assignment01$
```

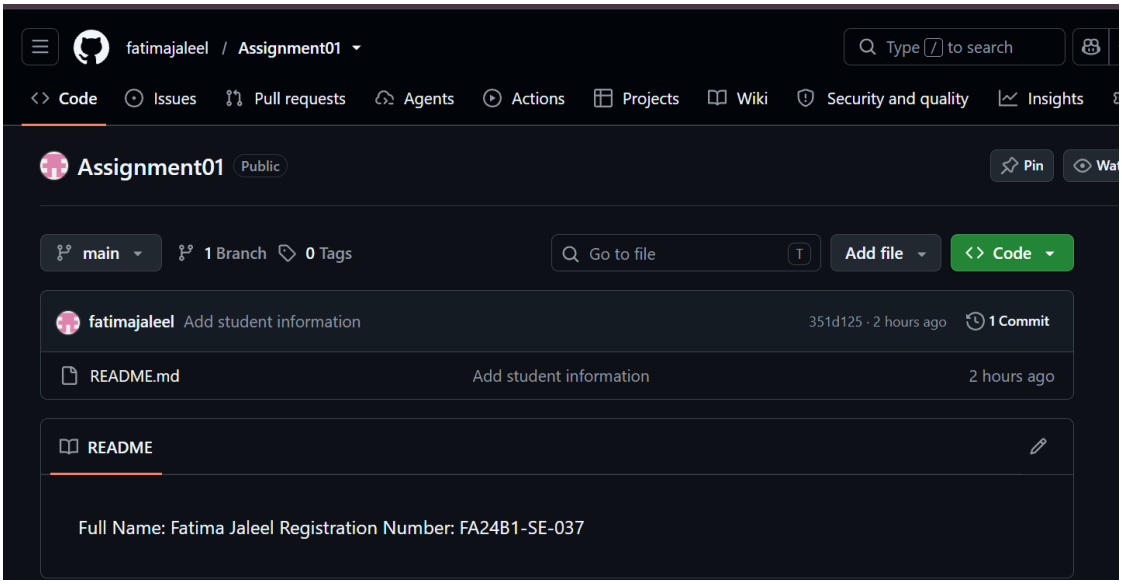
Step 2.2: Push to GitHub

Terminal output showing the successful execution of git push -u github main. Confirms branch tracking setup and initial object upload from the Ubuntu local repository to the GitHub remote repository.

```
fatimajaleel@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': fatimajaleel
Password for 'https://fatimajaleel@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/fatimajaleel/Assignment01.git/'
fatimajaleel@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': fatimajaleel
Password for 'https://fatimajaleel@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 273 bytes | 273.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/fatimajaleel/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
fatimajaleel@ubuntu:~/Assignment01$
```

Step 2.3: Verify the GitHub Repository

Web browser view of the public GitHub Assignment01 repository. Confirms repository ownership, public status, and rendered README.md displaying student identification info.



Task 3: Track Three Large Files with Git LFS

Step 3.1: Configure Git LFS

Initialization and configuration of Git LFS on the Ubuntu server. Displays output for git lfs install, git lfs version, tracking rule setup git lfs track "*.bin", and the resulting .gitattributes file.

```
fatimajaleel@ubuntu:~/Assignment01$ cd ~/Assignment01
fatimajaleel@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
fatimajaleel@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
fatimajaleel@ubuntu:~/Assignment01$ git lfs track "*.bin"
Tracking "*.bin"
fatimajaleel@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
fatimajaleel@ubuntu:~/Assignment01$
```

Step 3.2: Create and Verify Three LFS Files

Verification of large binary file tracking via Git LFS. Shows git lfs ls-files --size confirming that three created dummy files (large-file-1.bin, large-file-2.bin, large-file-3.bin) are tracked and exceed 100 MiB each.

```
fatimajaleel@ubuntu:~/Assignment01$ cd ~/Assignment01
fatimajaleel@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
fatimajaleel@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
fatimajaleel@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
fatimajaleel@ubuntu:~/Assignment01$ git add .gitattributes laarge-file-1.bin large-file-2.bin laarge-file-3.bin
fatal: pathspec 'laarge-file-1.bin' did not match any files
fatimajaleel@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
fatimajaleel@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
fatimajaleel@ubuntu:~/Assignment01$
```

Step 3.3: Commit and Push the LFS Files

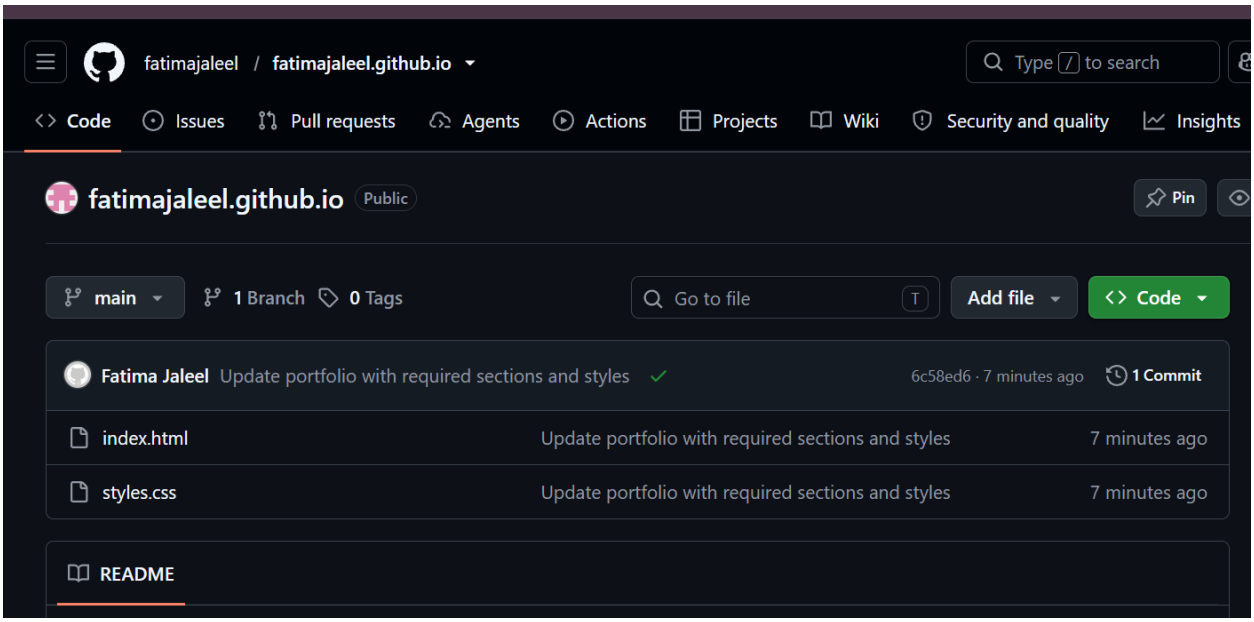
Terminal output of git push github main demonstrating Git LFS object uploading. Captures completed 100% (3/3) LFS pointer upload and successful commit push to GitHub.

```
fatimajaleel@ubuntu:~/Assignment01$ git commit -m "Add three files using Git LFS"
[main 483ec31] Add three files using Git LFS
 4 files changed, 10 insertions(+)
 create mode 100644 .gitattributes
 create mode 100644 large-file-1.bin
 create mode 100644 large-file-2.bin
 create mode 100644 large-file-3.bin
fatimajaleel@ubuntu:~/Assignment01$ git push github main
Uploading LFS objects: 100% (3/3), 338 MB | 714 KB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 796 bytes | 159.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/fatimajaleel/Assignment01.git
   d887f77..483ec31  main -> main
fatimajaleel@ubuntu:~/Assignment01$
```

Task 4: Create a Portfolio or CV with GitHub Pages

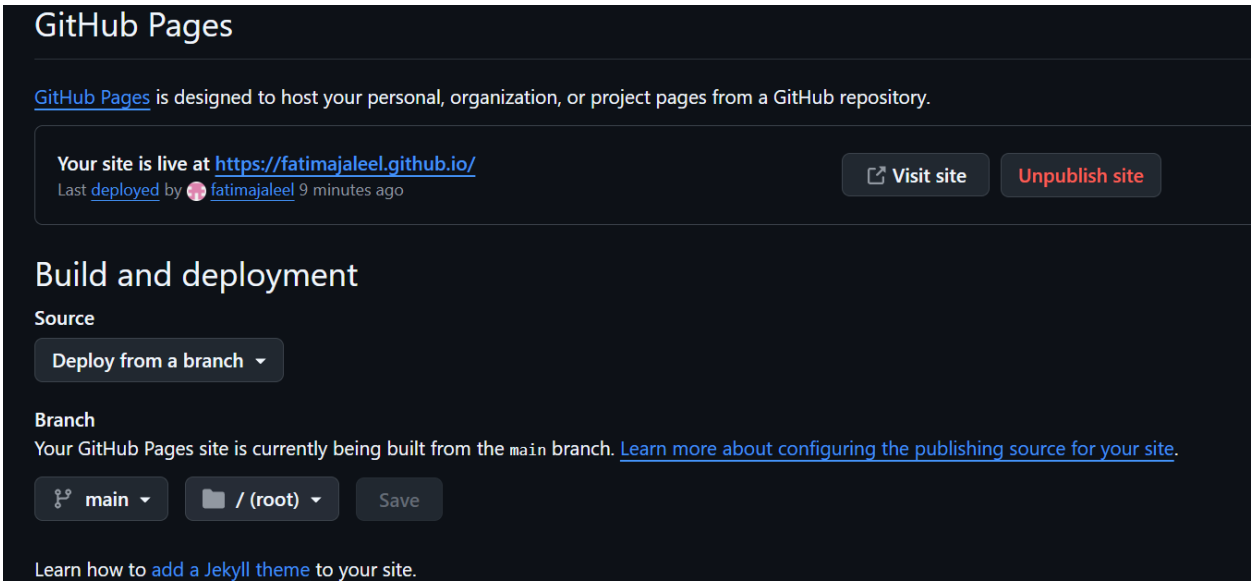
Step 4.1: Create and Push the Portfolio Repository

Web browser view of the public <github-username>.github.io repository on GitHub. Confirms the presence of committed website source files, including index.html and styles.css.



Step 4.2: Publish with GitHub Pages

GitHub Pages deployment status in repository settings (Settings -> Pages). Confirms build deployment from the main branch root folder and displays the active live site status banner.



Step 4.3: Verify the Live Portfolio

Rendered portfolio website loaded in browser at https://<github-username>.github.io/. Displays complete browser URL bar alongside required portfolio sections (**About Me**, **Education**, **Skills**, **Projects**).

Fatima Jaleel

Registration No: FA24B1-SE-037 | Software Engineering Student

About Me

Welcome to my portfolio! I am a Software Engineering student passionate about cloud computing and web technologies.

Education

BS Software Engineering - FA24B1-SE-037

Skills

- Git & GitHub Version Control
- HTML5 & CSS3
- Linux Administration & VMware
- Docker & Gitea

Projects

- **Assignment 01:** Local Gitea Server & Dual Remote Syncing
- **GitHub Pages Portfolio:** Live responsive personal portfolio